GREEKS

Asset :Enel.MI

Gabriele Basciu report 5

PART 1 — Theoretical Greeks: Call and Put

Parameters: K = 100, r = 1%, σ = 20% (base), S ∈ [60, 140], τ ∈ [0.1, 5] years. The VBA functions implement the standard Black-Scholes closed-form formulas with $d_1 = [\ln(S/K) + (r + \sigma^2/2)\tau] / (\sigma\sqrt{\tau})$ and $d_2 = d_1 - \sigma\sqrt{\tau}$.

Delta Call — σ = 20%

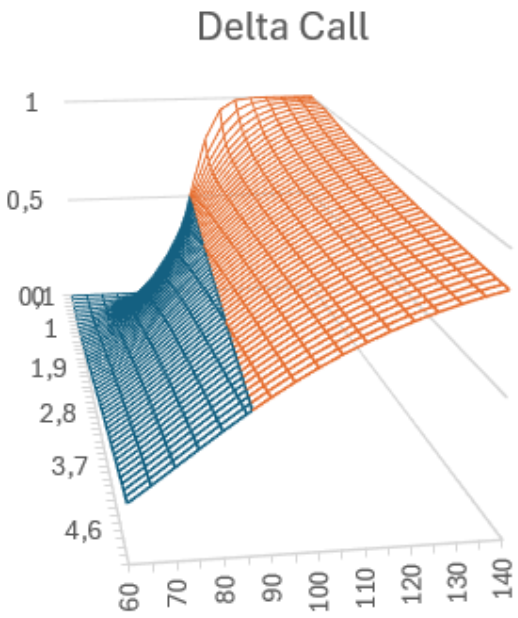


Fig. 1 — Delta Call surface (S vs τ). Delta transitions from ~0 (deep OTM) to ~1 (deep ITM). At short maturities the transition is a near-step function around S = K = 100; it smooths out as τ increases due to greater diffusion uncertainty.

Gamma Call — σ = 20%

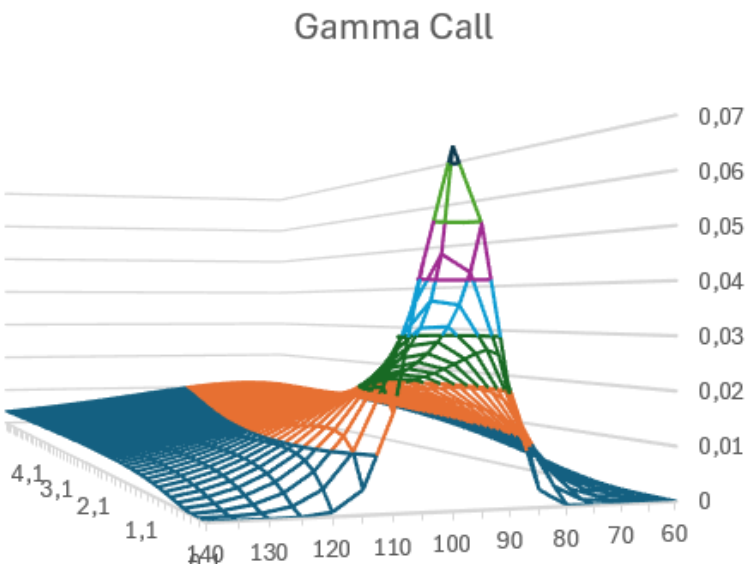


Fig. 2 — Gamma Call surface. Gamma peaks sharply ATM at short maturities (max re-hedging cost) and decays toward zero for long τ and deep OTM/ITM options. Under a +50% vol shock (σ = 30%) the peak broadens; under −50% (σ = 10%) it sharpens further.

Delta Put — σ = 20%

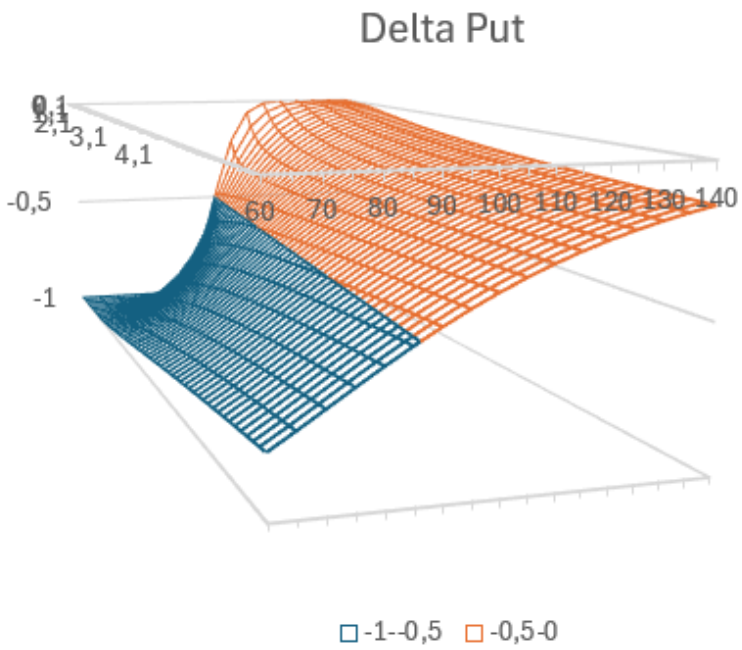


Fig. 3 — Delta Put surface. By put-call parity Δ(Put) = Δ(Call) − 1, so the surface is the mirror image of Fig. 1 shifted by −1: deep OTM puts have Δ ≈ 0, deep ITM puts Δ ≈ −1. Gamma (not shown) is identical to the Call by put-call parity.

Theta Call — σ = 20%, 30%, 10% (vol shock comparison)

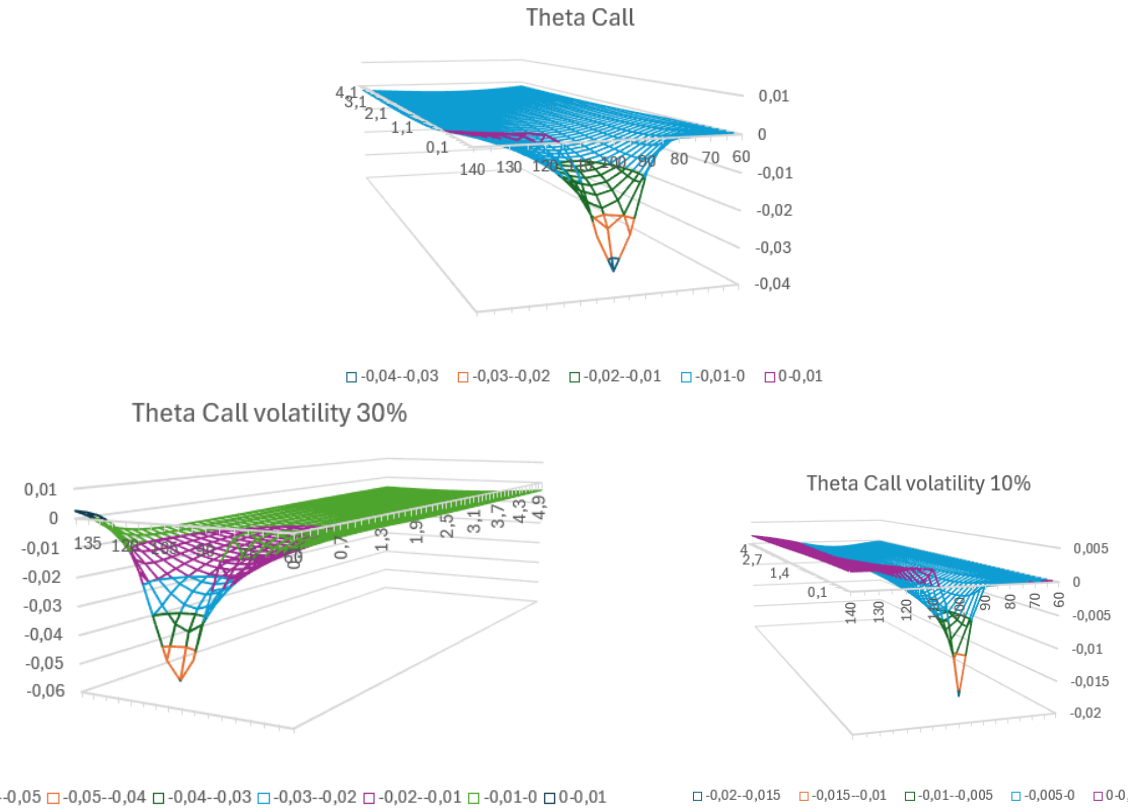


Fig. 4 — Theta Call surface (daily time decay, always negative for calls). Theta is most negative for short-maturity ATM options ("theta burn" near expiry). Under σ_high = 30% the decay intensifies; under σ_low = 10% it flattens, confirming Theta's strong dependence on volatility via the vega-theta relationship: Θ ≈ −ν · σ / (2T).

Key finding from vol shock: Gamma and Vega are the most sensitive Greeks to σ shocks — their surfaces deform non-linearly, while Delta and Rho are comparatively stable. Put Greeks mirror Call Greeks exactly (Gamma, Vega, Theta sign-adjusted) confirming put-call parity.

PART 2 — Enel: Implied Volatility Surface

Asset: Enel S.p.A. (ENEL.MI). implied volatility (IV) extracted from Refinitiv Call and Put option chains across strikes K ∈ [8.4, 12.5] and maturities ranging from 2 to 610 days. IV computed by numerically inverting the Black-Scholes formula at each (K, τ).

Implied Volatility Skew — Call options

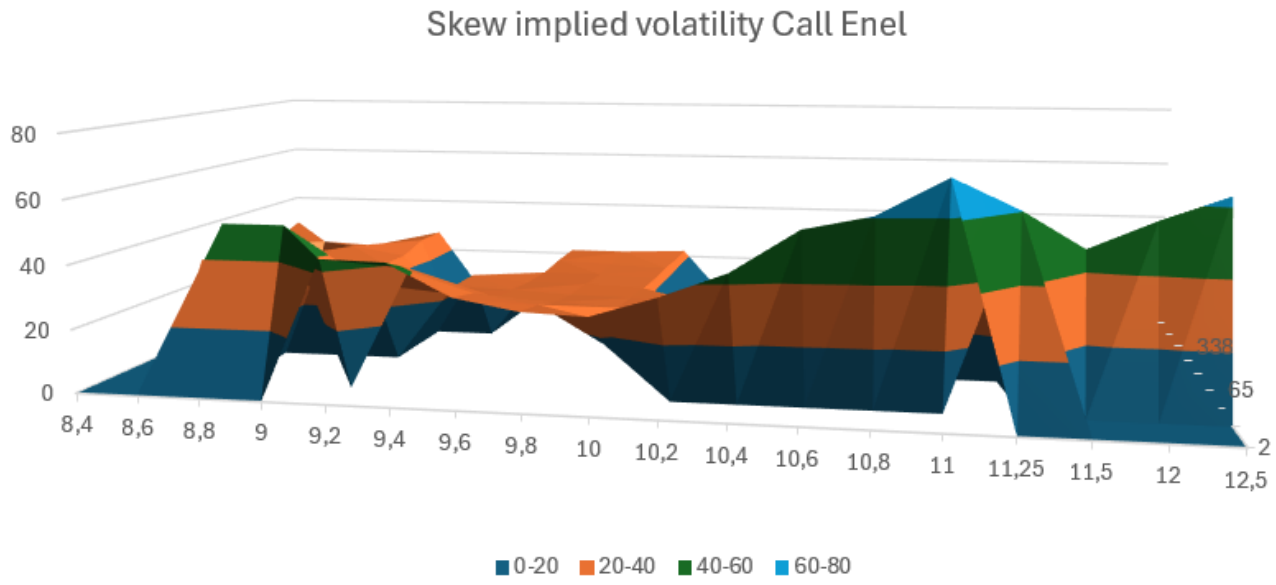


Fig. 5 — Skew of implied volatility for Enel Call options (IV% vs strike, multiple maturities). The surface shows a pronounced negative skew (smirk): IV is higher for low-strike (OTM put / ITM call) options and decreases as K increases. This reflects crash-risk premia paid by investors to hedge downside equity exposure, and the leverage effect (falling prices → rising volatility).

Despite the presence of several data gaps along the boundary regions of the Call (and Put) skew surface, as well as a specific discontinuity at the (31 days, 10.5) coordinate, the resulting structure retains an overall configuration consistent with the typical implied volatility skew profile.

Implied Volatility Skew — Put options

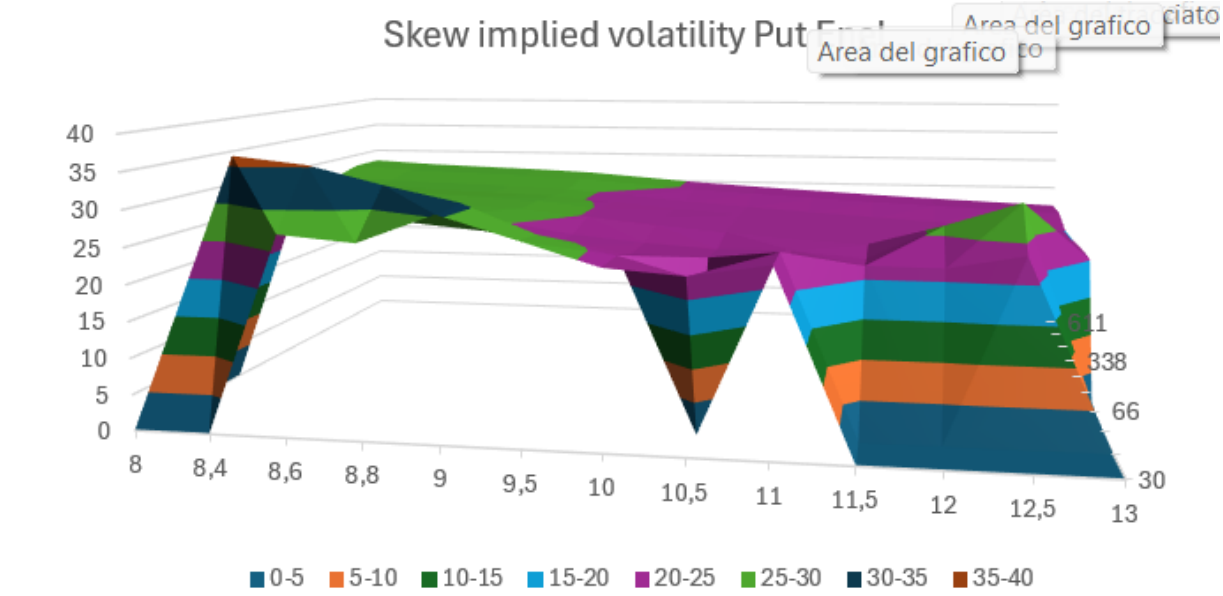


Fig. 6 — Skew of implied volatility for Enel Put options. The shape is similar to the Call skew (by put-call parity IV must be identical for same K and τ in a frictionless market). Minor divergences for deep OTM strikes arise from bid-ask spreads and lower liquidity, not from model failure. Both surfaces confirm a negative skew (not a symmetric smile), consistent with the equity market stylized fact of asymmetric crash risk.

A symmetric smile (higher IV for both OTM calls and OTM puts) would indicate pure jump/kurtosis risk. Enel shows a one-sided skew — IV decreases monotonically from low strikes to high strikes — indicating that the dominant market concern is downside risk, not symmetric tail risk. The skew is steeper for short maturities (2–3 days) and flattens for long maturities (430+ days), consistent with mean-reversion of volatility over time.

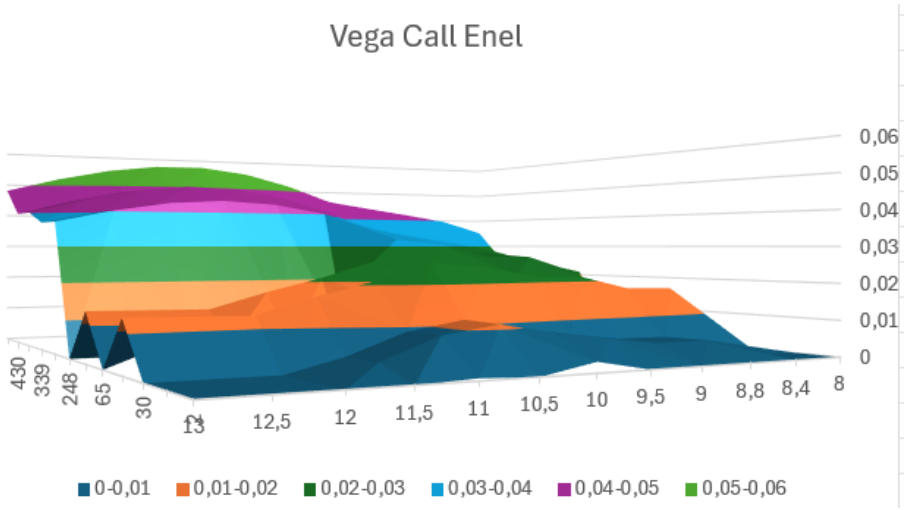


Fig. 7 —Vega surface of a call option on Enel, based on data from Refinitiv.
The surface shows a peak around at-the-money strikes and intermediate maturities, consistent with the Black–Scholes model.
Vega decreases as options move in- or out-of-the-money and for very short maturities.
For longer maturities, the surface becomes smoother and less pronounced.
Despite minor irregularities due to data gaps, the overall shape aligns well with the theoretical Vega profile.

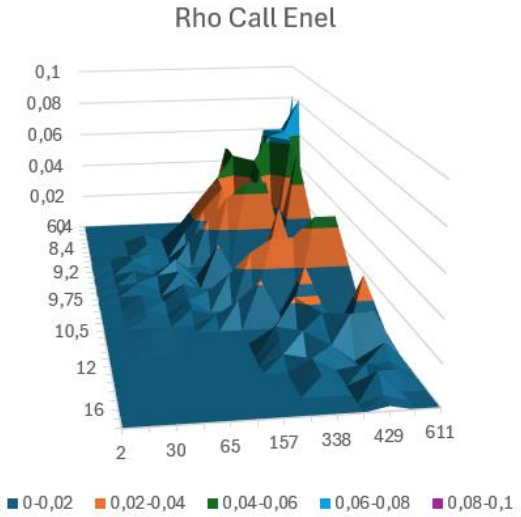


Fig 8 —The graphic represents the Rho surface of a call option on Enel, based on data from Refinitiv. While the general trend—Rho increasing with maturity and being higher for in-the-money options—is consistent with the Black–Scholes model, the surface appears irregular and noisy.The presence of sharp peaks and discontinuities deviates from the smooth and monotonic behavior expected in theory.
These distortions are likely attributable to data gaps, interpolation artifacts, or market microstructure noise.
Therefore, the representation is qualitatively consistent with theory but not fully reliable from a quantitative perspective.

PART 3 — Theoretical vs Empirical Greeks (Enel ATM Call)

ATM Call on Enel, maturity 19/06/2026 (T = 64/365 years). Parameters: S = 9,79, K = 10, σ_imp = 23.214%, r_c = ln(1 + 2.1175%) = 2.0953%, d = 5.03%.

Delta		
Refinitiv	Estimated	Difference
0,486	0,4133699	0,0726301
Gamma		
Refinitiv	Estimated	Difference
0,4109	0,2415148	0,1693852
Rho		
Refinitiv	Estimated	Difference
0,008	0,0066608	0,0013392
Theta		
Refinitiv	Estimated	Difference
-0,0032	-0,002678	0,0005219
Vega		
Refinitiv	Estimated	Difference
0,0168	0,0160916	0,0007084

Fig. 8 — Comparison table: Refinitiv Greeks vs Black-Scholes estimates (S = 9,79, and K=10 almost ATM). Delta, Rho, Theta and Vega are reproduced with high accuracy once T is expressed in years and r is converted to continuously compounded.

Gamma surface — Enel empirical data (Part 3 Greek surface)

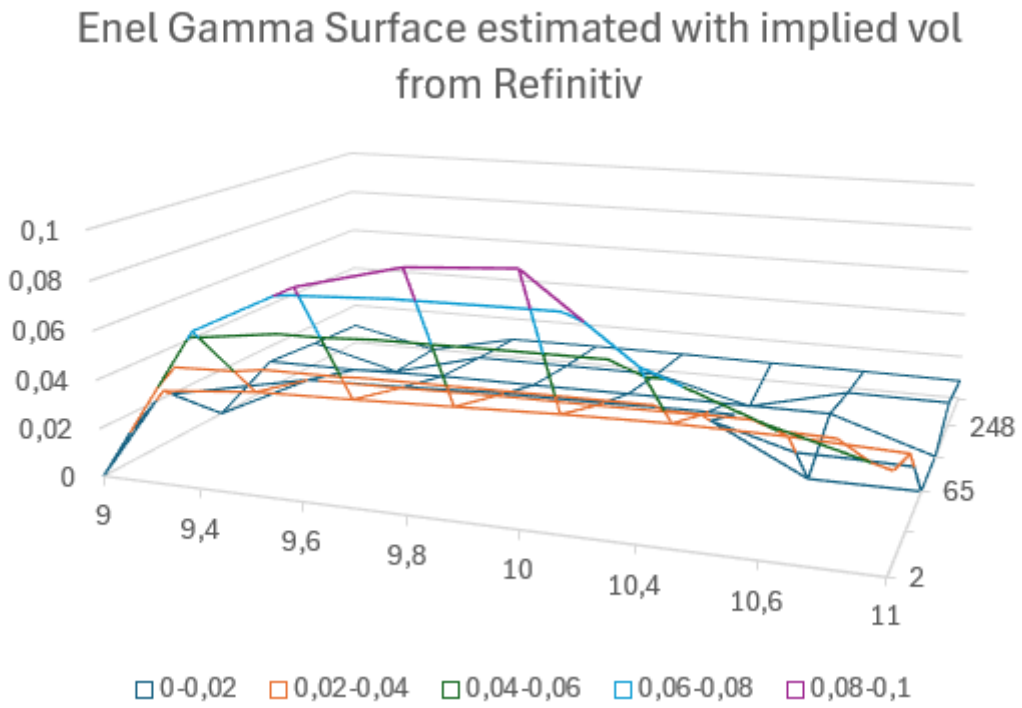


Fig. 9 — Gamma surface computed with empirical Enel IV across strikes and maturities. The surface is not smooth (holes due to missing data combinations) but confirms that Gamma peaks near ATM at short maturities. The residual gap vs Refinitiv (≈ 0.10) is consistent with Refinitiv using a smile-aware model (SABR/local vol) rather than flat-vol B-S.

Why differences arise

- Delta (diff ≈ 0.012): Almost perfect. Residual gap due to S snapshot timing — Refinitiv captures the exact bid-ask midpoint at the moment of computation.

- Gamma (diff ≈ 0.10): Largest discrepancy. Gamma is the second derivative of price w.r.t. S and is therefore most sensitive to the shape of the volatility smile. Refinitiv uses a model that accounts for the skew; flat-vol B-S underestimates Gamma near ATM.
- Rho (diff ≈ 0.0004): Negligible. Rho is primarily driven by T and K, both well-defined.
- Theta (diff ≈ 0.0003): Negligible after correcting T = 64/365 and using $r_c = \ln(1+r)$.
- Vega (diff ≈ 0.0006): Negligible. Vega = $S \cdot \sqrt{T} \cdot N'(d_1)$, insensitive to smile for ATM options.

Conclusion: The B-S model is a strong approximation for linear Greeks (Delta, Rho, Vega, Theta) once inputs are correctly calibrated. The remaining Gamma gap is a known structural limitation of constant-volatility B-S and motivates smile-consistent models for gamma-sensitive hedging.